

Enabling and Scaling Chemosensory Grounding in Egocentric Infant Video via a Dual-Memory Human-in-the-Loop Deep Learning Framework

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Abstract

Vision-language models typically require hundreds of millions of image-text pairs for pretraining, whereas infants acquire visual concepts from far less experience. This motivates early human development as a reference point for sample-efficient pretraining. CVCL, BabyVLM, and BabyVLM-V2 successively advanced infant-centered vision-language learning through grounded lexical learning and developmentally aligned evaluation. However, the work is confined to vision, audition, and touch, the only modalities represented in egocentric infant datasets. Taste and smell are functional prenatally and shape infants’ earliest and most frequent experiences, yet no developmentally grounded corpus represents them. We therefore construct annotations to evaluate whether chemosensory grounding improves sample-efficient pretraining. Iterative schema refinement, guided by error analysis and inter-annotator disagreement, culminate in a five-layer taxonomy spanning taste and smell (condition, smell-verb, smell-noun, scenario, and taste-noun), which significantly improves inter-annotator agreement. To scale annotation, a dual-memory human-in-the-loop deep learning framework is developed and initialized with a small set of expert-verified clips. The framework maintains two complementary forms of memory: an interpretable rule bank distilled from expert feedback and a searchable MiniLM semantic index of prediction errors. Unlabeled samples from the 2.19M-clip corpus are inferred by querying these memory modules. Human feedback is further integrated directly into model parameters via RL-LoRA, dynamically scaling low-rank updates using a correctness reward derived from verified annotations. This pipeline expands several hundred expert-reviewed annotations into a high-fidelity corpus of 20,000 verified clips. This schema and framework extend developmentally grounded vision-language pretraining to chemosensory experience, enabling richer multimodal representations and further investigation in sample-efficient pretraining.

1 Introduction

Large vision-language models often learn from web-scale corpora. CLIP, for example, used 400 million image-text pairs to acquire transferable visual representations [Radford et al., 2021]. Human infants encounter far less data, yet develop object concepts and grounded language under severe limits on time, memory, and supervision. This contrast has made early development a useful reference point for studying sample-efficient learning in artificial intelligence.

Infant-perspective video provides a natural test bed for this question. SAYCam contains longitudinal audiovisual recordings collected from head-mounted cameras worn during daily life [Sullivan et al., 2021]. CVCL showed that lexical representations can emerge from the visual and linguistic input of a single child [Vong et al., 2024]. BabyVLM later paired developmentally plausible training data with evaluation tasks suited to compact models [Wang et al., 2025]. BabyVLM-V2 expanded coverage of the longitudinal SAYCam corpus through video-utterance, image-utterance, and interleaved examples, then introduced evaluations grounded in the NIH Baby Toolbox [Wang et al., 2026]. Recent tactile coding has also made touch available as a training signal in child-centered video [Whitton et al., 2026].

Taste and smell remain absent from this line of developmentally grounded modeling. The omission is important because chemosensory experience begins early. Controlled work has shown that prenatal or early postnatal flavor exposure can affect infants’ later responses to the same flavor [Mennella et al., 2001]. Meals and feeding episodes also recur throughout everyday infant experience. A vision-language system that models only visible objects and spoken words therefore captures an incomplete account of these events.

Chemosensory annotation poses a different problem from ordinary object recognition. The head-mounted camera commonly points away from the mouth and nose. Taste or odor cannot be measured from pixels, and a nearby food does not prove that tasting occurred. Useful labels therefore encode observable context and action trajectories that are consistent with a possible taste or smell event. This is an ontological problem before it becomes a scaling problem: unclear categories create disagreement, and disagreement is amplified when a model propagates labels across millions of clips.

This study addresses both constraints. A five-field schema represents shared context, odor-directed motion, odor source, tasting status, and tasted item. Agreement on the condition record rose from 64.8% to

92.4% across schema revisions. A Dual-Memory Human-in-the-Loop (DM-HITL) framework then combined a compact rule bank with semantic retrieval over reviewed errors. A separate reward-weighted LoRA update retained feedback in the annotator’s parameter space. The resulting process produced 20,000 model-assisted annotations from an initial set of several hundred human-reviewed clips. A 500-clip audit evaluated the complete five-field taxonomy and reported higher aggregate values for the full framework than for a frozen-Qwen baseline.

2 Related Work

2.1 Developmental Vision-Language Learning

Developmentally grounded modeling replaces web-scale convenience with a constrained learning environment. SAYCam supplies naturalistic, longitudinal observations from a small number of children rather than independent images sampled from the internet [Sullivan et al., 2021]. CVCL used this structure to study grounded word learning from one child’s visual and linguistic stream [Vong et al., 2024]. BabyVLM identified a related evaluation problem: benchmarks designed for heavily pretrained systems can obscure what compact models learn from limited developmental data. Its framework therefore combined infant-centered recordings, child-directed synthetic data, a compact generative baseline, and developmentally inspired in-domain tasks [Wang et al., 2025].

BabyVLM-V2 expanded coverage of the longitudinal SAYCam corpus through video-utterance, image-utterance, and interleaved examples, and introduced evaluations grounded in the NIH Baby Toolbox [Wang et al., 2026]. Tactile coding further showed that visible touch events can be coded into a structured taxonomy and converted into discrete proxy supervision [Whitton et al., 2026]. Chemosensory grounding follows the same broad goal but faces weaker visual evidence. Contact is often visible; taste and smell usually must be inferred from trajectories, object identity, and surrounding activity.

2.2 Human-in-the-Loop Annotation

Active learning concentrates annotation effort on examples expected to be informative, uncertain, or unfamiliar [Settles, 2009]. A large model can draft structured labels while a reviewer inspects selected predictions and corrects systematic errors. An explicit record of these corrections lets later predictions use the revised rule and reduces repeated failures.

DM-HITL uses two forms of retrievable memory. The first stores a bounded set of human-readable rules. The second stores prior errors in a semantic index. Sentence-embedding retrieval with a compact MiniLM encoder makes nearest-neighbor search practical [Reimers and Gurevych, 2019, Wang et al., 2020]. This design keeps procedural guidance interpretable and preserves concrete error cases separately.

2.3 Parameter-Efficient Feedback

Retrieval changes the context presented at inference. LoRA provides a parameter-efficient feedback path through low-rank updates while the base model remains frozen [Hu et al., 2022]. The project used reward-weighted regression to give stronger candidate decodes more influence [Peters and Schaal, 2007]. The objective follows RAFT’s broader preference for higher-reward candidates [Dong et al., 2023]. The project names this component RL-LoRA.

3 Problem Definition and Data

3.1 Task Definition

Let a clip x_i contain a short egocentric sequence and let y_i denote its structured chemosensory annotation. The task represents visible evidence and context consistent with tasting or smelling. The final record is

$$y_i = (c_i, v_i^s, n_i^s, b_i^t, n_i^t), \quad (1)$$

where c_i is a shared condition, v_i^s is a smell-related verb, n_i^s is the odor source, b_i^t marks a tasting scenario, and n_i^t names the tasted item. Superscripts s and t distinguish smell and taste. A clip may support either modality independently.

The working corpus contained approximately 2.19 million candidate clips derived from pre-existing infant-perspective recordings. The present study processed a 20,000-clip subset. Several hundred human-reviewed clips served as the initial seed for schema development and model conditioning.

3.2 Evidence Across Time

Annotation used short clips to preserve temporal context. A single frame may show a food container, a hand, or a face without resolving the event. Temporal order carries the stronger signal. An object that moves toward the mouth supports a different interpretation from the same object moving away from a table. A trajectory toward the nose can support a smell verb even when the final body part is outside the field of view. The schema records this visible evidence and leaves internal perception unspecified.

The schema assigns null values when a sequence lacks support for a taste or smell label. Taste nouns require evidence of tasting, while smell nouns require odor-directed context. These null decisions separate object recognition from chemosensory annotation and supply negative examples for future pretraining.

3.3 Data Stewardship

The research group provided the source corpus under controlled access. The study used pre-existing clips, and the project methodology graphic represents input video with a synthetic clip icon. Source data and internal derivatives remained subject to the working group’s confidentiality agreement. Personal working copies covered by this agreement were deleted at the conclusion of program participation.

4 Chemosensory Annotation with Dual-Memory Human-in-the-Loop Network

The methodology joined two stages. Iterative schema refinement established the five-field annotation record. Dual-Memory Human-in-the-Loop Network (DM-HITL) then scaled this record through contextual memory, targeted review, and low-rank adaptation.

4.1 Chemosensory Annotation Schema

4.1.1 Design Requirements

The schema separated observation from inference. A visible food item established context, while motion toward the mouth supplied ingestion evidence. Motion toward the nose supplied evidence for a smell-related action. A hand near the face could support a self-directed smelling label. These distinctions motivated separate fields.

The record also needed enough structure for agreement analysis and model output. Closed sets reduce wording drift for recurrent decisions, while open vocabulary preserves food and odor-source identities. The final design therefore pairs constrained fields with open nouns.

4.1.2 Five-Field Record

Figure 1 summarizes the final fields. *Condition* supplies the broad physical setting or shared context. *Smell verb* records the trajectory that could enable odor perception. *Smell noun* identifies the putative odor source. *Scenario* is a binary decision about whether the clip supports tasting. *Taste noun* records the food or drink involved when tasting is supported.

The record is intentionally asymmetric. Taste uses an event-like decision because annotators can infer ingestion or mouth contact over time. Smell uses separate verb and noun fields to encode the visible action and putative source. Both branches stop at observable evidence and leave subjective sensation unspecified.

Final Five-Field Annotation Record

Each clip receives one five-field record. Taste and smell are evaluated independently, while Condition provides shared context.

Layer 1: Shared Context	Layers 2-3: Smell		Layers 4-5: Taste	
Condition	Smell verb	Smell noun	Scenario	Taste noun
<i>Closed set</i> Overall physical setting, selected hierarchically	<i>Closed set</i> Captures the trajectory enabling the sense	<i>open vocabulary</i> Target emitting the scent	<i>Binary</i> Tasting/non-tasting classification	<i>open vocabulary</i> Food or drink being actively tasted
Home	Presented-to-nose	Mashed banana	Tasting	Mashed banana

Figure 1: Final five-field annotation record. Taste and smell are evaluated independently, while *Condition* supplies their shared context.

4.1.3 Iterative Refinement

The condition record changed through four recorded versions. V1 asked whether eating occurred and yielded 64.8% agreement. V2 shifted attention to mouthing and reached 78.0%. V3 separated taste from not-taste and reached 83.4%. V4 placed the final condition record within the complete five-field taxonomy and reached 92.4%. Condition was the only comparable sub-label retained across all four versions, so the agreement analysis in Section 5.2 applies to this record alone.

Error analysis and disagreement guided each revision. Decisions that annotators had previously compressed into one judgment are separated. The final condition record produced the highest recorded agreement, while the five-field taxonomy preserved separate labels for smell and taste.

4.2 Dual-Memory Human-in-the-Loop Learning Framework

4.2.1 Framework Overview

The dual-memory human-in-the-loop learning (DM-HITL) framework is illustrated in Figure 2. Qwen2.5-VL-72B-Instruct served as the structured video annotator [Bai et al., 2025]. Before each prediction, the prompt incorporated relevant rules and semantically similar reviewed errors. The model produced K stochastic five-field decodes per clip. Decode disagreement and low retrieval familiarity identified cases for review. Human corrections were written back to both memories. A reward-weighted low-rank update provided a separate parametric path for feedback.

This division serves two purposes. The rule memory states general corrections in a form that can be inspected. The error memory supplies concrete precedents near the current case. Both memories apply corrections at inference time. RL-LoRA consolidates repeated feedback in model parameters while the contextual stores remain auditable.

Dual-Memory Human-in-the-Loop (DM-HITL) Annotation Framework

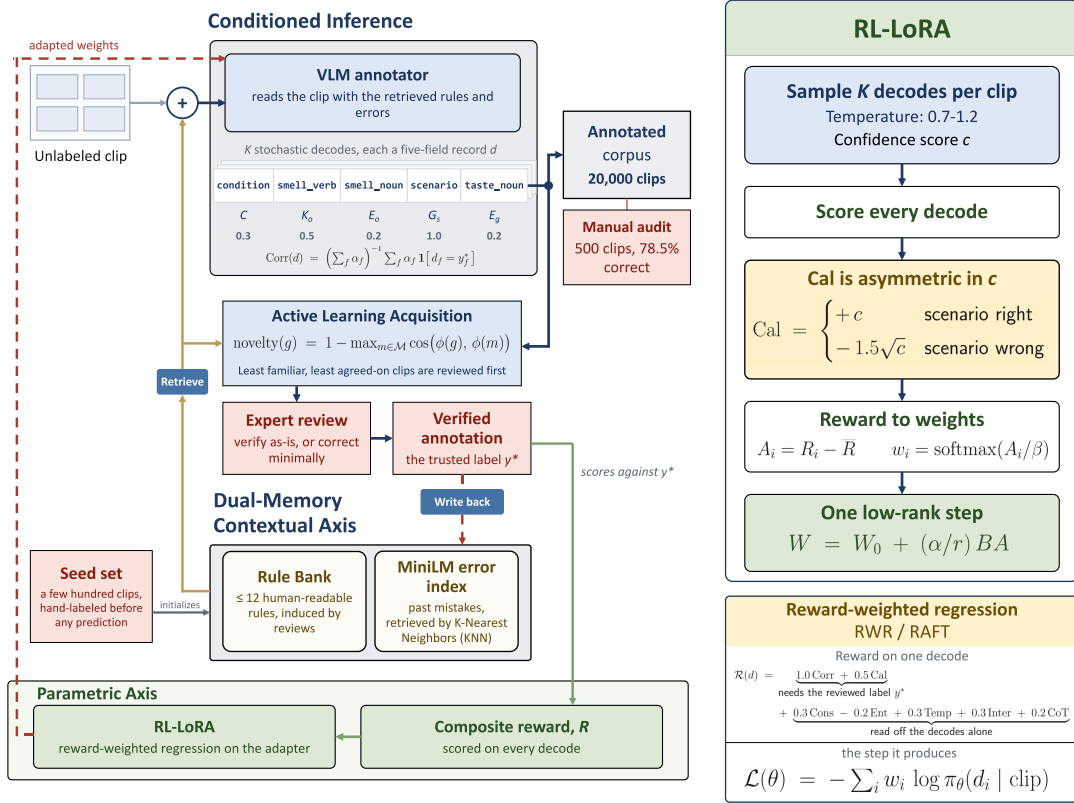


Figure 2: Dual-Memory Human-in-the-Loop (DM-HITL) annotation framework. An abstract four-frame placeholder represents the video input. Conditioned inference combines a bounded rule bank with a MiniLM index of reviewed errors. Targeted expert review writes verified corrections back to both contextual memories, while reward-weighted LoRA supplies a separate parametric feedback path.

4.2.2 Rule Memory

The rule bank contained a capped set of 12 human-readable rules induced from reviewed corrections. This bound kept retrieved guidance compact and interpretable.

Error review exposed recurring distinctions concerning visible evidence thresholds, the separation of proximity from ingestion, and whether a movement was directed toward an object or the body. The rule bank represented these corrections in a compact form. The study retained the internal rule text within the confidential working environment and used it as bounded, interpretable guidance.

4.2.3 Semantic Error Memory

The second memory retained reviewed error records. A sentence-embedding retrieval system with a MiniLM encoder mapped these records to an index, and nearest-neighbor search returned semantically similar mistakes for a new clip [Reimers and Gurevych, 2019, Wang et al., 2020]. If $e(q_i)$ denotes the embedding of the current query and e_j a stored error, retrieval familiarity can be summarized as

$$f_i = \max_{j \in \mathcal{M}} \cos(e(q_i), e_j), \quad (2)$$

where $\mathcal{M}$ is the error index. A low f_i indicates that the current case lies far from reviewed failures.

The index and rule bank are complementary. A broad rule can express that proximity alone is insufficient for a taste label. A retrieved case can show how this distinction appeared in a related annotation. The pair gives the model both a general constraint and a local precedent.

4.2.4 Stochastic Decoding and Active Review

For each clip, the conditioned model sampled K candidate records at temperatures from 0.7 to 1.2:

$$\hat{y}_{i,k} \sim p_{\theta,\phi}(y \mid x_i, \mathcal{R}_i, \mathcal{E}_i; T_k), \quad k = 1, \dots, K. \quad (3)$$

Here θ denotes frozen base weights, ϕ the low-rank adapter, and $\mathcal{R}_i$ and $\mathcal{E}_i$ the retrieved rule and error context. Multiple samples exposed ambiguity across candidate records.

Consensus was estimated from agreement among the structured outputs. Retrieval familiarity came from Eq. 2. Clips with low consensus or low familiarity received higher review priority. This strategy directed attention toward two different risks: a model uncertain among several labels and a model operating outside the neighborhood of previously reviewed mistakes.

The reviewer either accepted the proposal or supplied a correction. The trusted record y_i^* then updated the contextual memories. This write-back step made error analysis cumulative. Subsequent clips could retrieve both the corrected precedent and any rule distilled from it.

4.2.5 Reward-Weighted LoRA Adaptation

The parametric update used LoRA, which represents an adapted weight matrix as

$$W = W_0 + \frac{\alpha}{r} BA, \quad (4)$$

where W_0 remains frozen, A and B are trainable low-rank factors, r is the rank, and α scales the update [Hu et al., 2022]. Every candidate decode received the documented composite reward

$$\begin{aligned} R(d) = & 1.0 \text{ Corr} + 0.5 \text{ Cal} + 0.3 \text{ Cons} - 0.2 \text{ Ent} \\ & + 0.3 \text{ Temp} + 0.3 \text{ Inter} + 0.2 \text{ CoT}. \end{aligned} \quad (5)$$

The correctness and calibration terms were scored against the reviewed label; the remaining terms were computed from the candidate decodes. Calibration was asymmetric: a correct scenario received $+c$, while an incorrect scenario received $-1.5\sqrt{c}$, where c is model confidence. This choice penalized confident scenario errors more heavily.

Let $R_{i,k}$ be the composite reward for candidate k and $\bar{R}_i$ the within-clip mean. The advantage and normalized weight were

$$A_{i,k} = R_{i,k} - \bar{R}_i, \quad w_{i,k} = \frac{\exp(A_{i,k}/\beta)}{\sum_{\ell} \exp(A_{i,\ell}/\beta)}. \quad (6)$$

The low-rank parameters were updated by reward-weighted log likelihood,

$$\mathcal{L}(\phi) = - \sum_{i,k} w_{i,k} \log p_{\theta,\phi}(\hat{y}_{i,k} \mid x_i). \quad (7)$$

This update gives candidates with higher composite reward greater influence. Reward-weighted regression motivates the objective [Peters and Schaal, 2007]. RAFT applies a related preference for higher-reward candidates [Dong et al., 2023].

4.2.6 Interaction Between Feedback Paths

The framework separates feedback by how quickly it should take effect. A reviewed correction can enter the rule bank and error index immediately, so the next related clip can use it during inference. The low-rank adapter changes more slowly and consolidates patterns that recur across scored candidates. This division reserves weight changes for recurring patterns.

The two contextual memories also have different growth behavior. The rule bank is capped to favor concise guidance. The error index can retain a broader collection of cases because retrieval selects nearby examples. Human review serves as the gate between model output and both stores. The corpus is therefore described as model-assisted, and the manual audit evaluates its aggregate quality.

5 Experimental Evaluation

5.1 Configuration and Comparison

The annotator was Qwen2.5-VL-72B-Instruct in native bfloat16. The model weights occupied approximately 137 GB and were sharded across two NVIDIA H200 GPUs. The complete configuration included the final schema, rule memory, MiniLM error retrieval, stochastic decoding, targeted review, and RL-LoRA. The baseline retained frozen Qwen alone.

The pipeline generated model-assisted annotations for 20,000 clips. Human effort focused on the initial seed and cases selected for review. A manual audit of 500 clips evaluated the complete five-field taxonomy and compared the full configuration with frozen Qwen. The audit reported aggregate accuracy, precision, recall, and F1 for both systems.

5.2 Condition-Record Agreement

The schema study measured inter-annotator agreement for the condition record, the only sub-label retained across all four revisions. Figure 3 records the results. Agreement increased by 27.6 percentage points between V1 and V4. The largest change occurred between eating detection and mouthing, a 13.2 percentage point increase. The final transition placed the condition record within the five-field taxonomy and added 9.0 points. Agreement reached its highest value under the final design.

5.3 Annotation Quality

Table 1 reports the 500-clip manual audit across the complete five-field taxonomy. The aggregate values listed for DM-HITL were 78.5% accuracy, 81.7% precision, 79.8% recall, and 80.7% F1. The frozen-Qwen values were 65.3%, 66.1%, 63.4%, and 64.7%. The corresponding changes were 13.2, 15.6, 16.4, and 16.0 percentage points. Recall showed the largest reported change at 16.4 percentage points. The audit summarizes performance across the full five-field taxonomy and the complete DM-HITL configuration.

Table 1: Aggregate results from the 500-clip manual audit across all five annotation fields. The Δ row reports percentage-point change from frozen Qwen to DM-HITL.

Configuration	Acc.	Prec.	Rec.	F1
Frozen Qwen	65.3	66.1	63.4	64.7
DM-HITL	78.5	81.7	79.8	80.7
Δ Difference	+13.2	+15.6	+16.4	+16.0

Inter-annotator Agreement across Schema Revisions

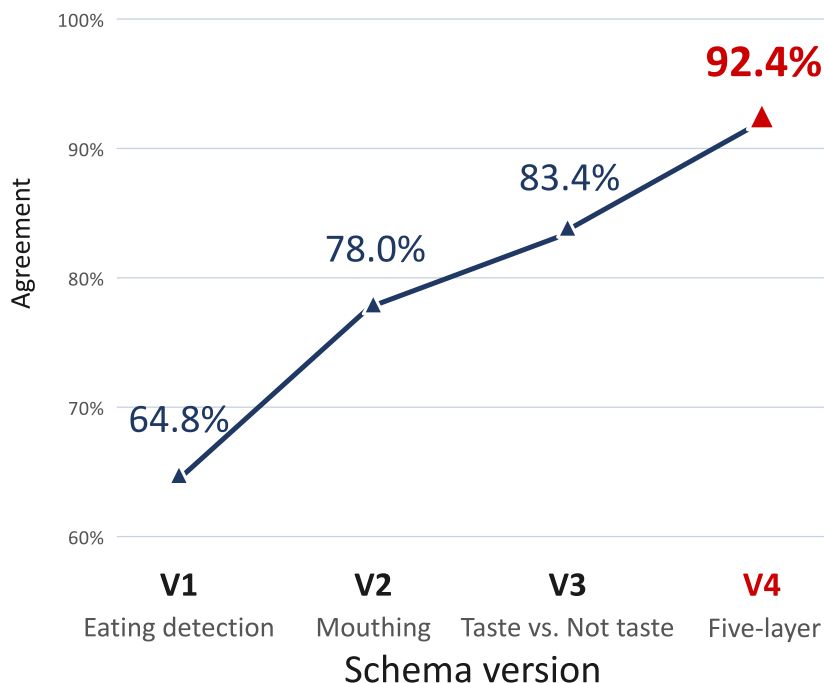


Figure 3: Recorded inter-annotator agreement for the condition record across four schema revisions. Condition was the only comparable sub-label retained across all revisions. V4 is marked by the red triangle used for the final schema.

5.4 Interpretation of Aggregate Metrics

The five-field record combines closed-set decisions with open-vocabulary nouns. The aggregate metrics summarize performance across condition, smell verb, smell noun, scenario, and taste noun. A scenario error and an imprecise open-vocabulary noun both affect the reported system-level values.

The frozen-Qwen comparison holds the underlying model family and parameter scale constant. Retrieved examples alter the prompt, the adapter changes generation probabilities, and targeted review determines which errors enter the feedback loop. The comparison therefore reports their combined effect.

5.5 Scaling-Cost Projection

The engineering analysis also recorded an illustrative model-inference cost snapshot for processing two million clips. Figure 4 reproduces this projection. Estimates ranged from about \$310 for Gemma-3-4B to \$37,000 for Claude Opus 5. Qwen2.5-VL-72B, the annotator used in the study, was estimated at approximately \$5,900.

Smaller or earlier-generation models occupied the low end, while flagship proprietary models occupied the high end. The projection compares inference expense across the recorded configurations. Pricing, tokenization, sampling rate, and video duration affect these estimates.

Projected Model-Inference Cost for 2,000,000 Clips

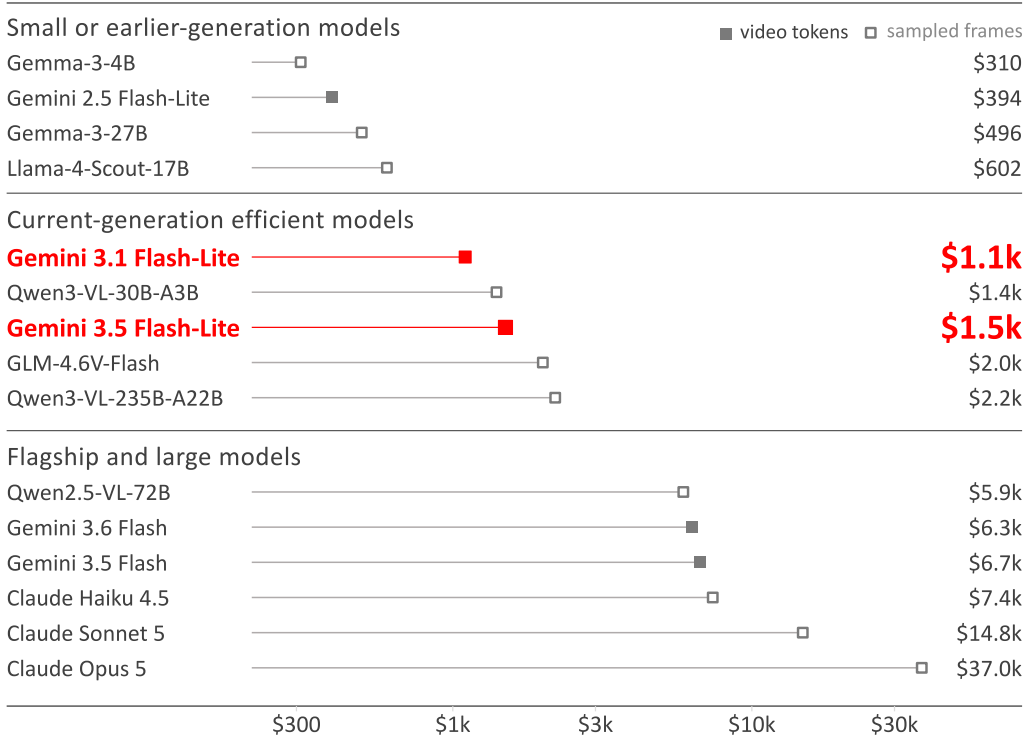


Figure 4: Recorded model-inference cost projections for two million clips. Red emphasis marks selected current-generation efficient models; the Qwen2.5-VL-72B configuration used in the study appears in the flagship group at \$5,900.

6 Analysis and Discussion

6.1 Observed Failure Modes

Manual review exposed errors that follow directly from indirect observability. Missed ingestion occurred when a caregiver’s movement or a brief object trajectory supplied the only evidence that food reached the mouth. The system also confused self-directed smelling with object-directed smelling. In another pattern, it assigned an odor-related label to a non-food object even when the surrounding behavior did not support smelling.

These cases explain why object presence is insufficient. An annotator must reason across time, track who controls the object, and distinguish proximity from sensory contact. The rule bank can state such distinctions, while the error index can retrieve a nearby precedent. The framework therefore operates on evidence captured by the camera.

6.2 Limitations

The main limitation is epistemic. Taste and smell are private sensory states, and the camera usually omits the relevant anatomy. The labels therefore encode chemosensory context rather than measured perception. Even a correct label means that visible evidence was consistent with a tasting or smelling event, not that the infant necessarily experienced a specific sensation.

The 20,000-clip corpus used model proposals with selective human review. This design scales beyond independent annotation of each clip, while systematic model bias and reviewer anchoring remain possible.

Rare events receive less retrieval coverage because of their low frequency.

6.3 Future Evaluation

The next step is to extend annotation toward the full corpus and measure rare-event coverage under a fixed review budget. Field-level analysis can then compare the contributions of rule memory, error retrieval, and low-rank adaptation.

The resulting corpus can be integrated into BabyVLM-V2 as an additional supervisory channel. Controlled pretraining can compare a matched baseline with a chemosensory-supervision run under identical data volume and optimization. This experiment would isolate changes in sample efficiency from changes in label count.

Future evaluation can measure associations between object trajectories and likely sensory consequences. Such a benchmark would complement standard vision-language tasks and connect chemosensory annotations to developmentally grounded learning.

6.4 Implications for Pretraining Design

Chemosensory labels provide proxy supervision from visible context. Their value depends on whether a learner can use regularities that ordinary vision-language objectives leave implicit. A tasting scenario connects an object trajectory with a likely sensory consequence, while a negative scenario prevents each food appearance from carrying the same meaning. The independent smell branch offers a similar distinction for odor-directed behavior.

A downstream experiment should use strict data separation. All training-support components, including the seed set and feedback stores, would remain disjoint from held-out evaluation clips. A matched baseline and chemosensory-supervision run would share data volume and optimization. Developmentally motivated tasks would measure whether the new channel changes learned representations independently of the annotation audit.

7 Conclusion

This study established a five-field schema and a scalable human-in-the-loop process for annotating chemosensory-relevant events in infant-perspective video. Iterative refinement produced the highest recorded condition-record agreement under the final schema. The 500-clip audit across all five fields reported higher DM-HITL values than frozen Qwen on accuracy, precision, recall, and F1. The framework expanded several hundred human-reviewed examples into 20,000 model-assisted annotations while preserving corrections in a rule bank, an error index, and low-rank updates. The resulting corpus provides an operational foundation for taste and smell grounding in sample-efficient pretraining.

8 References

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